



AI-Powered Collections Strategy

Leveraging Agentic AI for Scalable, Fair, and Effective Debt
Management at **Geldium**

How the System Works

Inputs

Customer repayment history, Month 4 payment status, credit utilization, debt-to-income ratio, credit score, loan balance, and account data.

Decision Logic

Conceptual Logistic Regression model estimates delinquency probability. Business rules add thresholds, escalation paths, and policy constraints.

Actions

Prioritize early review and proportionate outreach. Lower-risk cases receive standard reminders; higher-risk routed to trained collectors.

Learning Loop

Track repayment outcomes, intervention results, fairness metrics, and model performance. Use findings to continuously refine the system.

Role of Agentic AI

AUTONOMOUS

-  Monitor new data and calculate approved risk scores.
-  Prioritize cases and trigger standard, automated reminders.
-  Generate respectful, policy-approved outreach drafts.
-  Summarize account history and risk signals for review.
-  Log all actions and outcomes for compliance monitoring.

HUMAN OVERSIGHT

-  Review high-risk, ambiguous, or disputed cases.
-  Approve material collection actions and exceptions.
-  Handle hardship requests and personalized repayment plans.
-  Review overrides, complaints, and fairness alerts.
-  Approve model or rule changes before production use.

Responsible AI Guardrails

Fairness Checks

Monitor intervention rates and model errors across customer groups to prevent biased outcomes.

Explainability

Provide understandable reasons for risk flags using approved factors like payment behavior and DTI.

Regulatory Compliance

Protect financial information and maintain auditable records to align with ECOA, GDPR, and local laws.

Human-in-the-Loop

Keep human review for critical decisions; support manual overrides, appeals, and hardship handling.

Continuous Monitoring

Track performance, complaints, drift, and fairness post-deployment. Pause system if guardrails breach.

Expected Business Impact

Quantitative Outcomes (KPIs)

-  Reduced delinquency rate among monitored customers.
-  Increased early identification and intervention rates.
-  Improved collection recovery and repayment rates.
-  Enhanced collector productivity and reduced case time.
-  Optimized model performance (fewer false positives).

Qualitative Outcomes

-  Improved trust via earlier and proportionate support.
-  Enhanced fairness with consistent human review options.
-  Fewer unnecessary contacts and clearer communication.
-  Better, more accessible hardship and repayment support.
-  Increased scalability without sacrificing protections.

Pilot recommendation: Measure outcomes over an initial 3-month controlled pilot before scaling.

Executive Takeaway

- ✓ Use predictive risk signals to identify customers who may benefit from earlier intervention.
- ✓ Use agentic AI for repeatable, low-risk operational tasks while retaining human judgment.
- ✓ Build fairness, explainability, privacy, and compliance into the system from day one.
- ✓ Validate business and customer outcomes through a controlled pilot before scaling.

THE GOAL

A scalable collections capability that improves prioritization and efficiency without removing human judgment or customer protections.